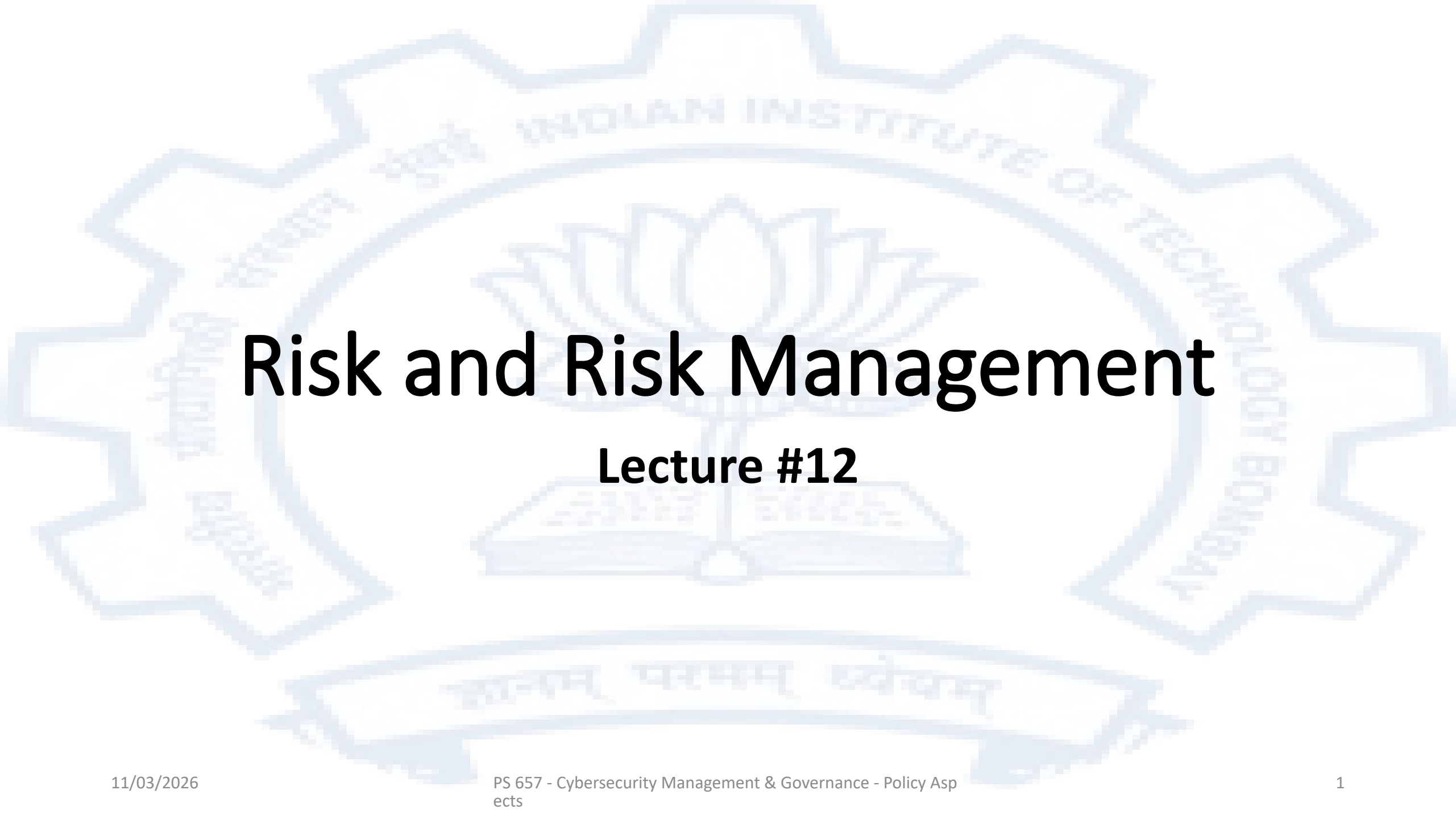


The background of the slide features a large, light blue watermark of the Vellore Institute of Technology (VIT) logo. The logo is a gear-like shape with a central emblem containing a lotus flower and an open book. The text "VIT VELLORE INSTITUTE OF TECHNOLOGY" is written around the top inner edge of the gear, and a Sanskrit motto is at the bottom.

Risk and Risk Management

Lecture #12

Risk – a working definition

- Risk is the possibility that human actions or events lead to consequences that have an impact on what humans value.
- Impact is the magnitude of harm that can be caused by actions or events
- The expected loss over a given time period is
 - Occurrences X impact per occurrence
- The occurrences depend on the likelihood or probability of occurrence per unit time multiplied by the time

What is it that distinguishes the thousands of years of history from what we think of as modern times? The answer goes way beyond the progress of science, technology, capitalism and democracy.

.....

The revolutionary idea that defines the boundary between modern times and the past is the mastery of risk: the notion that the future is more than a whim of the gods and that men and women are not passive before nature. Until human beings discovered a way across that boundary, the future was a mirror of the past the murky domain of oracles and soothsayers who held a monopoly over knowledge of anticipated events

- Peter L Bernstein, "Against the Gods: The remarkable story of risk", Wiley, 1996 & 1998

Risk in ancient history

- Events were thought to be ordained by divine forces and uncertainty regarding their occurrence constituted risk
- Experience offered some clues on how to navigate the uncertainties of weather, food availability etc and survive
- Various priests, astrologers, oracles and mystics were consulted on the likelihood of occurrence of events and what might be done to ensure or avoid those events
- These were the earliest instances of risk management

Homer, capturing memory, sang of Zeus, Hera, Athena, Apollo, and the corps of divinities responsible for the victory at Troy as well as the misadventures of Odysseus on his return home. But by 585 BC, the Greek philosopher Thales used his observations, written data, and deductions to predict an eclipse of the sun, even though he continued to profess a belief in these gods. A century later Herodotus used intelligent “enquiry” to write “history,” but he too persisted with the power of divinities. It was finally Thucydides, in the early 400s BC, who proposed a “new penetrating realism,” one that “removed the gods as explanations of the course of events.” Thucydides was “fascinated by the gap between expectation and outcome, intention and event.” Perhaps he should be called the father of risk management.

[\(PDF\) A Brief History of Risk Management \(researchgate.net\)](#)

The influence of the scientific method

- Several advances as part of the overall development of the scientific method have significantly improved our ability to quantify risk
- Some of the most important of these
 - The discovery and systematic cataloguing of the laws of nature so that there was a more sophisticated understand of causality as opposed to correlation
 - The development of the theory of probability
- These developments meant that scientific tools could be used to predict the likelihood of events and thereby achieve a finer quantification of risk

Probability

how likely something is to happen

- Oxford dictionary

- Can the likelihood be inferred through causality
 - A shoots B. B will probably die.
- Should it be estimated statistically
 - A fair die can display 6 possibilities. The probability of a specific outcome is $1/6$
- Is it uncertain
 - Enough is not known to make a rational estimate

The evolution of probability

- Leonardo Pisano (who introduced Arabic numerals)
- Luca Paccioli (double-entry bookkeeping)
- Girolamo Cardano (measuring the probability of dice)
- Blaise Pascal (“fear of harm ought to be proportional not merely to the gravity of the harm, but also to the probability of the event”)
- John Graunt (who calculated statistical tables)
- Daniel Bernoulli (the concept of utility)
- Jacob Bernoulli (the “law of large numbers”)
- Abraham de Moivre (the “bell” curve and standard deviation)
- Thomas Bayes (statistical inference)
- Francis Galton (regression to the mean)
- Jeremy Bentham (the law of supply and demand)

Risk and uncertainty

Uncertainty must be taken in a sense radically different from the familiar notion of Risk from which it has never been properly separated.

*...It will appear that a **measurable** uncertainty, or “risk” proper is so far different from an **unmeasurable** one that it is not in effect an uncertainty at all.*

- Knight, Frank H., 1964. Risk, Uncertainty & Profit. New York: Century Press

Some Methods of dealing with societal risk

- Insurance
 - Insurance dates back at least to the time of the Mesopotamian civilization
 - This was well before the development of probability and so outcomes even till the mediaeval period were erratic
- Common Law
 - Compensation for negligence
- Direct Government Intervention
 - Examples are legislation in safety, public health, pollution and transport regulation
- Private Sector Self Regulation
 - Examples are Standards and Professional certifications

Important shifts in risk management from earlier times to the present

- Change in the nature of risks
- Increased life expectancy
- Emergence of new risks
- Improved ability to measure risk
- Increase in numbers of specialist studying risk
- Increased use of formal risk analysis techniques and frameworks
- Increase in the participation of special interest groups
- Increasing role of government
- Increasing demands for protection

Increased awareness at the Organizational and Individual level

- There is an increased awareness of the need to manage risk both at the organizational and individual levels
 - Enterprise risk management is an established discipline
 - There is a great deal of focus on personal risk
 - Investment advice
 - Periodic medical screening
 - Lifestyle changes
 - Personal fitness and wellness
 - Insurance

Risk Assessment

Risk assessment involves

- Identification of a risk
- Estimation of its likelihood
- Estimation of its impact
- A method of combining likelihood with impact to arrive at a categorization of the risk (in terms of severity)
- Three common categories of risk (in terms of severity) are
 - Intolerable – Requires abandoning of the system or reduction of risk via countermeasures
 - Tolerable – represents a level of risk that has been reduced to as low as reasonably possible (ALARP) or as low as reasonably allowable (ALARA)
 - Acceptable

Quantifying Risk

- Likelihood can be estimated probabilistically if a certain degree of knowledge is available
- Impact must be assessed independently
- The combination of likelihood over a given period with the impact can yield a “loss expectancy” for the period
- There are several combinations possible
 - Low likelihood, low impact
 - Low likelihood, high impact
 - High likelihood, low impact
 - High likelihood, high impact

Risk assessment

- Identify factors that contribute to the occurrence of an event
- Identify how the presence, or absence or occurrence or non-occurrence of these factors contribute to the likelihood of occurrence of the event
- Identify the impact of a single occurrence of the event and quantify that in appropriate units of “harm”
 - Money lost
 - Time lost
 - People impacted
 - Etc
- The risk assessment may be presented as risk metrics

Risk treatment/mitigation

- Once risk has been assessed it needs to be treated/mitigated. The expected loss may be within or beyond the risk tolerance limit for the specific situation
- In general there are three accepted methods of risk treatment
 - Risk avoidance
 - Risk transference
 - Risk acceptance
- Should the breaching of a risk tolerance limit be considered equivalent to the event actually occurring?

Risk Management

Risks themselves may have differing nature and based upon that different types of management plans may be needed. For example one way of further categorizing risk is

- Routine risks
- Complex risks
- Uncertain risks
- Ambiguous risks

Enterprise Risk Management Frameworks

- There are very many enterprise risk management frameworks in use
- The most widely used ones are
 - COSO ERM Framework – This is quite general
 - ISO 31000 Risk Management Standard – This is quite general
 - COBIT – Focusses on IT systems
 - NIST Cyber Security Framework – Focusses on cyber security
 - ITIL Service Lifecycle – Focusses on IT and IT based service delivery
 - OCTAVE Allegro – Developed by CMU's SEI this specifically focusses on operational risk

Categorizing Enterprise Risks

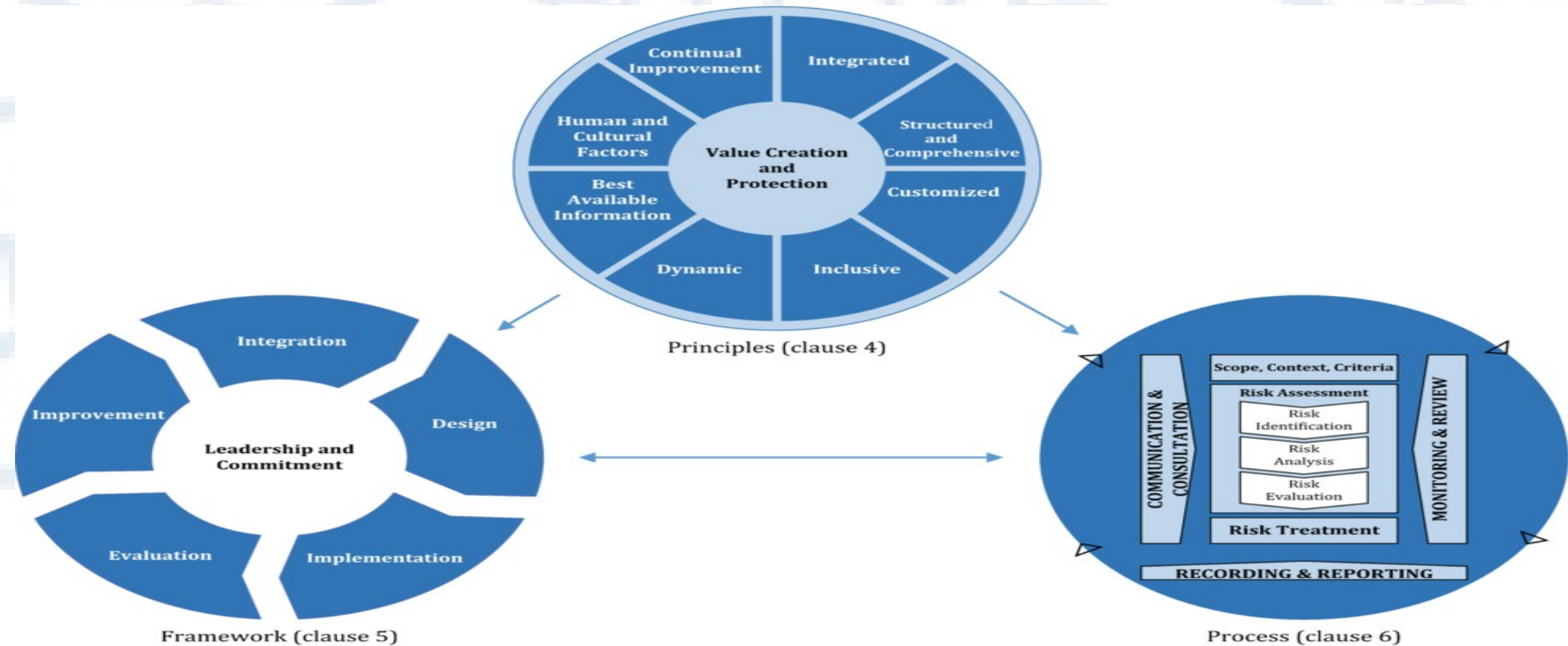
- Enterprise Risks are commonly grouped into categories
- Some common categories are
 - Financial Risks
 - Operational Risks
 - Strategic Risks
 - Legal/Regulatory/Compliance Risks
 - Reputational Risks

The COSO ERM Framework

The COSO framework has five major components.

- Control Environment
- Risk Assessment
- Control Activities
- Information and Communication
- Monitoring

ISO 31000 Risk Management Standard



Risk avoidance

Risk avoidance means taking steps to counter the risk so that the level of risk is reduced. Some example are

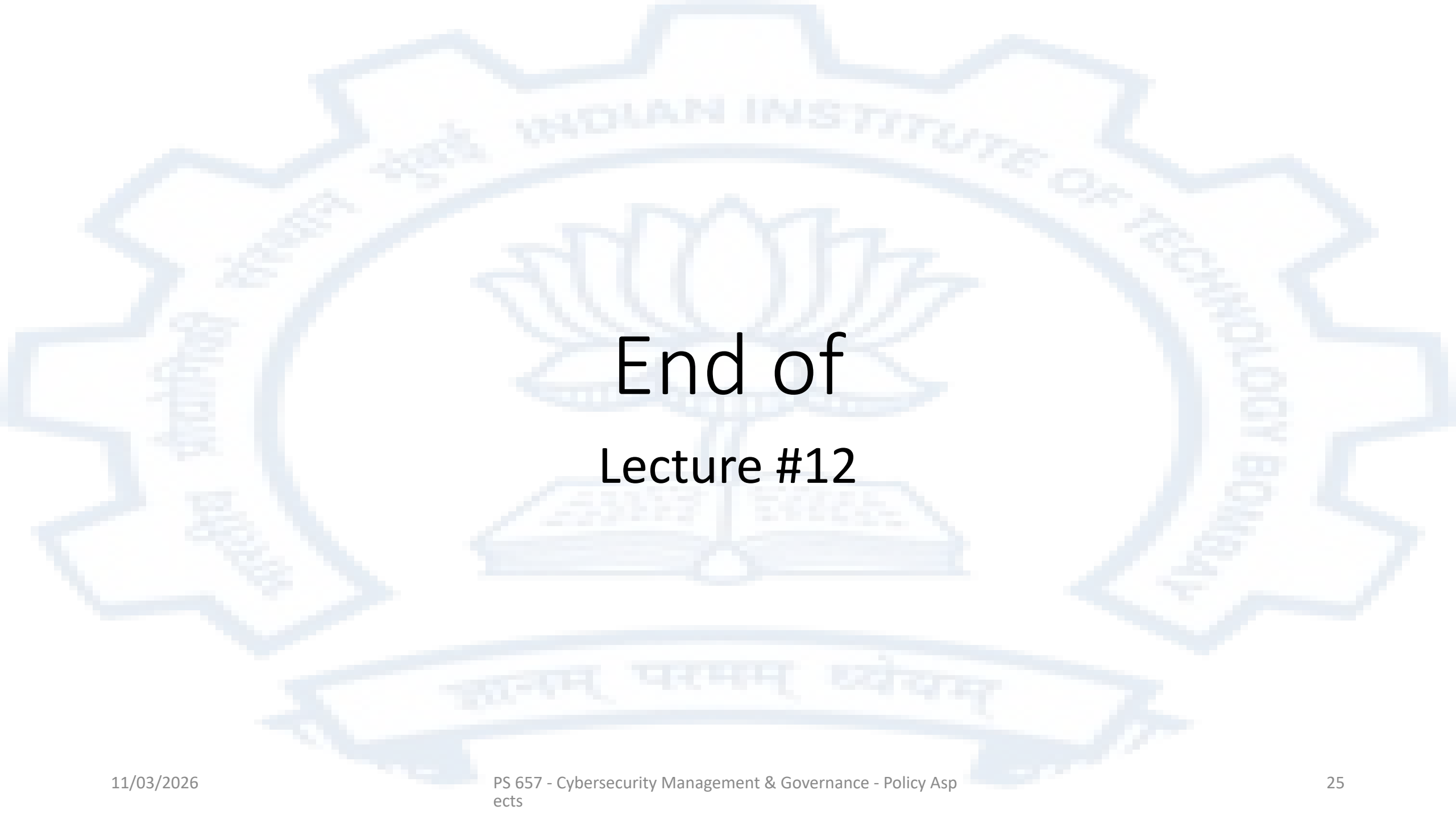
- Putting a lock on a door
- Putting on a seat belt
- Spreading investment across different investment options

Risk transference

Risk transference involves transferring the risk to another party usually based on some consideration. The primary example is insurance.

Summary

- Risk is the likelihood and impact of an event happening
- To deal with risk it must first be realistically assessed and quantified
- Once a risk has been assessed it must be treated
- The three most important ways to treat risk are
 - Risk avoidance
 - Risk transference
 - Risk acceptance
- An important policy question is should the breaching of a risk tolerance level be considered as equivalent to the event having occurred?

The background of the slide features a large, light blue watermark of the Indian Institute of Technology Bombay logo. The logo is circular with a gear-like outer border. Inside the border, the text "INDIAN INSTITUTE OF TECHNOLOGY BOMBAY" is written in a circular path. In the center of the logo is a stylized lotus flower with five petals, resting on a base that resembles an open book. Below the lotus, there is a banner with text in Devanagari script.

End of Lecture #12